AC444
VALUATION
AND SECURITES ANALYSIS

Week 9
Estimating the Cost of Capital

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ESTIMATING THE COST OF EQUITY CAPITAL

- To estimate a firm's cost of equity capital, one could use the CAPM:

$$r_{Ei} = r_F + \beta_{Ei}[r_M - r_F]$$

with:

- ✓ r_{Ei} denoting the cost of equity capital of firm i;
- ✓ β_{Ei} denoting the sensitivity of stock i to the market portfolio;
- ✓ r_F denoting the risk-free rate;
- ✓ r_M denoting the cost of capital of the market portfolio.

ESTIMATING THE COST OF EQUITY CAPITAL

- To estimate the β of listed stock i, one uses the market model:

$$r_{Eit} = \alpha + \beta_{Ei}\, r_{Mt} + \varepsilon_{it}$$

with:

- ✓ r_{Eit} denoting the monthly (or daily) return of stock i on day t;
- ✓ r_{Mt} denoting the monthly (or daily) return of a relevant large index (proxy for the market portfolio) on day t.

ESTIMATING THE COST OF EQUITY CAPITAL

- To estimate the β of listed stock i, one uses the market model:

$$r_{Eit} - r_{Ft} = \alpha + \beta_{Ei}\left(r_{M,t} - r_{Ft}\right) \quad + \varepsilon_{it}$$

with:

- ✓ r_{Eit} - $r_{F,t}$ denoting the monthly (or daily) excess return of stock i over the risk-free return on day t;
- ✓ r_{Mt} - $r_{F,t}$ denoting the monthly (or daily) return of a relevant large index (proxy for the market portfolio) over the risk-free return on day t.

in which case, α is Jensen's α.

ESTIMATING THE COST OF EQUITY CAPITAL

- As betas tend to mean-revert, raw betas from the use of the market model are smoothed. Bloomberg uses the following process:

$$\text{Adjusted beta} = 0.33 + 0.67\ (\text{Raw Beta})$$

- This process of estimating equity betas is however quite imprecise (as it involves large standard errors).

ESTIMATING THE COST OF EQUITY CAPITAL

- More precise estimates of securities' betas can be obtained by recognizing that:

  - ✓ the unlevered beta of a stock only reflects the operating risk (risk of the net operating assets) of the firm ;

  - ✓ The unlevered beta should be the same regardless of the firm owning the net operating assets.

ESTIMATING THE COST OF EQUITY CAPITAL
Using Industry Betas

- Assume you wish to to compute the equity beta of stock i, which may or may not be listed, go through the following sequence:

1. Use the market model and the smoothing formula to compute the equity betas of all stocks in the same industry;

2. Unlever the betas of these stocks (using stocks' leverage ratios) to get estimates of the beta of the industry's net operating assets;

3. Your best estimate of the beta of the industry's net operating assets (unlevered beta) is thus the average estimate which can be computed from step 2;

4. Lever the average estimate of the unlevered beta using stock i's leverage ratio to generate the equity beta of stock i;

LEVERING AND UNLEVERING THE COST OF EQUITY CAPITAL

Where it all starts from

- According to Modigliani and Miller, the market value of a firm's economic assets, net operating assets V_U and tax shields V_{TS}, should be equal to the market value of its financial claims such as debt V_D and equity V_E:

$$V_U + V_{TS} = EV = V_D + V_E \qquad (1)$$

- It follows that the risk of the company's economic assets, operating assets and tax shields, must be equal to the risk of the financial claims over these assets:

$$\frac{V_U}{V_U+V_{TS}}(r_U) + \frac{V_{TS}}{V_U+V_{TS}}(r_{TS}) = \frac{V_D}{V_D+V_E}(r_D) + \frac{V_E}{V_D+V_E}(r_E) \qquad (2)$$

LEVERING AND UNLEVERING THE COST OF EQUITY CAPITAL

Where it all starts from

- Using the same reasoning and process for betas than for costs of capital, we obtain:

$$\frac{V_U}{V_U+V_{TS}}(\beta_U)+\frac{V_{TS}}{V_U+V_{TS}}(\beta_{TS})=\frac{V_D}{V_D+V_E}(\beta_D)+\frac{V_E}{V_D+V_E}(\beta_E) \qquad (3)$$

- and:

$$\beta_E=\beta_U+\frac{V_D}{V_E}(\beta_U-\beta_D)-\frac{V_{TS}}{V_E}(\beta_U-\beta_{TS}) \qquad (4)$$

LEVERING AND UNLEVERING THE COST OF EQUITY CAPITAL

- Most companies manage their debt-to-enterprise ratios to stay within a small range;

- In this case, we could assume that debt-to-enterprise value remains constant;

- This implies that:

  - ✓ $r_{TS} = r_U$
  - ✓ $\beta_{TS} = \beta_U$

LEVERING AND UNLEVERING THE COST OF EQUITY CAPITAL

Debt Remains a Constant Proportion of Enterprise Value

- Equation (2) thus leads to the following relations between the cost of capital (beta) of the unlevered firm, cost (beta) of debt, and the cost (beta) of equity:

$$r_U = \frac{V_D}{V_D + V_E} r_D + \frac{V_E}{V_D + V_E} r_E$$

$$r_E = r_U + \frac{V_D}{V_E}(r_U - r_D)$$

$$\beta_U = \frac{V_D}{V_D + V_E} \beta_D + \frac{V_E}{V_D + V_E} \beta_E$$

$$\beta_E = \beta_U + \frac{V_D}{V_E}(\beta_U - \beta_D)$$

LEVERING AND UNLEVERING THE COST OF EQUITY CAPITAL

Debt Remains a Constant Proportion of Enterprise Value

- If debt is furthermore risk-free, then:

$$\beta_E = \left(1 + \frac{V_D}{V_E}\right)\beta_U$$

LEVERING AND UNLEVERING THE COST OF EQUITY CAPITAL

Another Special Case

- Some financial analysts set the required rate of return on interest tax shields equal to the cost of debt:
  - ✓ $r_{TS} = r_D$
  - ✓ $\beta_{TS} = \beta_D$

LEVERING AND UNLEVERING THE COST OF EQUITY CAPITAL

Risk of the Tax Shields Being Equal to the Risk of Debt

- It then follows from Equation (2) that:

$$\frac{V_U}{V_U + V_{TS}}(r_U) + \frac{V_{TS}}{V_U + V_{TS}}(r_D) = \frac{V_D}{V_D + V_E}(r_D) + \frac{V_E}{V_D + V_E}(r_E)$$

- and:

$$V_U\, r_U + V_{TS}\, r_D = V_D\, r_D + V_E\, r_E$$

$$V_U\, r_U = (V_D - V_{TS})\, r_D + V_E\, r_E$$

LEVERING AND UNLEVERING THE COST OF EQUITY CAPITAL

Risk of the Tax Shields Being Equal to the Risk of Debt

- It then follows that:

$$r_U = \frac{V_D - V_{TS}}{V_D - V_{TS} + V_E} r_D + \frac{V_E}{V_D - V_{TS} + V_E} r_E \qquad (5)$$

$$r_E = r_U + \frac{V_D - V_{TS}}{V_E}(r_U - r_D) \qquad (6)$$

LEVERING AND UNLEVERING THE COST OF EQUITY CAPITAL

Risk of the Tax Shields Being Equal to the Risk of Debt

- Similarly:

$$\beta_U = \frac{V_D - V_{TS}}{V_D - V_{TS} + V_E}\beta_D + \frac{V_E}{V_D - V_{TS} + V_E}\beta_E$$

$$\beta_E = \beta_U + \frac{V_D - V_{TS}}{V_E}(\beta_U - \beta_D)$$

LEVERING AND UNLEVERING THE COST OF EQUITY CAPITAL

Risk of the Tax Shields Being Equal to the Risk of Debt

- Further assume that debt is constant in nominal terms;

- If T_{CM} denotes the marginal corporate tax rate, it then follows that:

$$V_{TS} = \frac{V_D\, r_D\, T_{CM}}{r_D} = V_D\, T_{CM}$$

LEVERING AND UNLEVERING THE COST OF EQUITY CAPITAL

Risk of the Tax Shields Being Equal to the Risk of Debt

- It then follows from Equations (5) and (6) that:

$$r_U = \frac{V_D(1 - T_{CM})}{V_D(1 - T_{CM}) + V_E} r_D + \frac{V_E}{V_D(1 - T_{CM}) + V_E} r_E$$

$$r_E = r_U + \frac{V_D(1 - T_{CM})}{V_E}(r_U - r_D)$$

LEVERING AND UNLEVERING THE COST OF EQUITY CAPITAL

Risk of the Tax Shields Being Equal to the Risk of Debt

- Similarly:

$$\beta_U = \frac{V_D(1 - T_{CM})}{V_D(1 - T_{CM}) + V_E}\beta_D + \frac{V_E}{V_D(1 - T_{CM}) + V_E}\beta_E$$

$$\beta_E = \beta_U + \frac{V_D(1 - T_{CM})}{V_E}(\beta_U - \beta_D)$$

LEVERING AND UNLEVERING THE COST OF EQUITY CAPITAL

Risk of the Tax Shields Being Equal to the Risk of Debt

- If debt is furthermore risk-free, then:

$$\beta_U = \frac{\beta_E}{\left[1 + (1 - T_{CM})\frac{V_D}{V_E}\right]}$$

$$\beta_E = \left[1 + (1 - T_{CM})\frac{V_D}{V_E}\right]\beta_U$$

LEVERING AND UNLEVERING THE COST OF EQUITY CAPITAL

Summary: Unlevered Cost of Capital

	Fluctuating Debt	Constant Debt
$r_{TS} = r_U$	$r_U = \frac{V_D}{V_D + V_E} r_D + \frac{V_E}{V_D + V_E} r_E$	
$r_{TS} = r_D$	$r_U = \frac{V_D - V_{TS}}{V_D - V_{TS} + V_E} r_D + \frac{V_E}{V_D - V_{TS} + V_E} r_E$	$r_U = \frac{V_D(1 - T_{CM})}{V_D(1 - T_{CM}) + V_E} r_D + \frac{V_E}{V_D - V_{TS} + V_E} r_E$

LEVERING AND UNLEVERING THE COST OF EQUITY CAPITAL

Summary: Levered Cost of Equity

	Fluctuating Debt	Constant Debt
$r_{TS} = r_U$	$r_E = r_U + \frac{V_D}{V_E}(r_U - r_D)$	
$r_{TS} = r_D$	$r_E = r_U + \frac{V_D - V_{TS}}{V_E}(r_U - r_D)$	$r_E = r_U + (1 - T_{CM})\frac{V_D}{V_E}(r_U - r_D)$

LEVERING AND UNLEVERING THE COST OF EQUITY CAPITAL

Summary: Levered Beta

	Fluctuating Debt	Constant Debt	Risk-Free Debt
$r_{TS} = r_U$	$\beta_E = \beta_U + \frac{V_D}{V_E}(\beta_U - \beta_D)$		$\beta_E = \left(1 + \frac{V_D}{V_E}\right)\beta_U$
$r_{TS} = r_D$	$\beta_E = \beta_U + \frac{V_D - V_{TS}}{V_E}(\beta_U - \beta_D)$	$\beta_E = \beta_U + \frac{V_D}{V_E}(1 - T_{CM})(\beta_U - \beta_D)$	$\beta_E = \left[1 + (1 - T_{CM})\frac{V_D}{V_E}\right]\beta_U$